

Valeria Mascolo

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About

Finalist of the [French complex Systems Society PhD Award 2025](#). Deeply passionate about climate physics, statistical and dynamical properties of rare and extreme climate events and tipping points, disordered and complex systems. Profoundly intrigued and concerned about the societal, economical, environmental and health impacts of climate change. Coordinator of the [EArth](#) project: bridging art and climate by reimagining art masterpieces under the effects of climate change. Enthusiastic about art history, baking, swimming and running.

Professional experience

- since 2025 – **Research Scientist, National Centre for Atmospheric Science, University of Reading**
Simulating, forecasting and associated impacts of tipping points: Subpolar Gyre and Greenland Ice Sheet, as part of PROMOTE, advised by Prof. Reinhard Schiemann and Dr. Andrea Dittus.
- 2020–2021 **Trainee at the European Commission's Joint Research Centre, Text Mining & Analysis Competence Centre team**, Machine learning to analyze social media sentiment on climate engagement, focusing on the European Green Deal.

Education

- 2021–2024 **Ph.D. in Physics, Ecole Normale Supérieure de Lyon**
Title: *Studying extreme heatwaves using novel theoretical approaches*, advised by Prof. Freddy Bouchet. **Maria Skłodowska-Curie ITN H2020 fellow**, within project [EDIPI](#). Published manuscript available [here](#).
- 2019–2020 **M.Sc. in Complex systems modelling, King's College London, Highest honors**
Dissertation: *Analysis of Bayesian classification - Naive Bayes approach for the analysis of medical data*
Selected modules: Statistical learning - Pattern Recognition, neural Networks and Deep Learning - Statistics in finance - Dynamical and equilibrium analysis of complex systems - Theory of complex networks.
- 2016–2019 **B.Sc. in Physics Engineering, Politecnico di Torino, Highest honors**
Dissertation: *Deterministic and stochastic analysis of epidemics. SIS and SIR approaches to model epidemics*.
Selected modules: Advanced mathematical methods - Linear algebra and geometry - Mechanics and thermodynamics - Quantum, Nuclear & Solid-state physics - Computer science - Physics of complex systems.

Research visits

- Oct. – Dec. 2023 **Visiting PhD student of Prof. Francesco Ragone, Royal Meteorological Institute of Belgium**
- Oct. – Dec. 2022 **Visiting PhD student of Prof. Nili Harnik, Tel Aviv University, Department of Geoscience**

Outreach and volunteering

- since 2025 – **EGU's Climate: Past, Present & Future and Nonlinear processes in Geosciences ECS division member**, Co-chief blog editor, events and meetings organiser.
- since 2025 – **Climate Ambassador**, Member of [Climate Ambassadors](#), to support nurseries, schools and colleges to build their climate action plans.

Participation to schools and trainings

- 2025– **Impacts of climate change on health courses**
 - South and Southeast Asia Climate and Health Responders, Columbia University.
 - European Climate Resilient Health Systems Course, Columbia University.

2021–2024 **Third academic year (2023–2024)**

- EDIPI ITN 5th training week at KIT, Karlsruhe, Germany
- EDIPI ITN 6th training week - *Network-wide ESR Research and Soft Skills Training* at Cyprus Institute, Nicosia, Cyprus
- Participation to the European Geosciences Union General Assembly, Wien, Austria
- Workshop - *Prévisibilité dans les sciences de l'atmosphère et Points de bascule* for the Group De Recherche "Défis théoriques pour les sciences du climat", Institut Henri Poincaré, Paris, France
- Workshop - *Interfaces dans le système climatique*, for the Group De Recherche "Défis théoriques pour les sciences du climat", Grenoble, France

Second academic year (2022–2023)

- **Workshop** - *Mathematics and theoretical physics for climate dynamics*, Ecole Normale Supérieure de Lyon, Lyon, France
- **Summer school** - *Climate sensibility*, Ecole Normale Supérieure de Lyon, Lyon, France
- EDIPI ITN 4th training week - *Idealized modelling in climate sciences*, Institut Royal Météorologique, Brussels, Belgium
- **Summer school** - *Out of equilibrium statistical physics*, Beg Rohu, France
- Course - *Granular, jammed and disordered media*, Ecole Normale Supérieure de Lyon, Lyon, France
- Participation to the European Geosciences Union General Assembly, Wien, Austria

First academic year (2021–2022)

- Course - *Large deviations in physics*, Ecole Normale Supérieure de Lyon, Lyon, France
- Course - *Climate change, climate change impacts, and energetic transition*, Ecole Normale Supérieure de Lyon, Lyon, France
- **Summer school** - *Artificial intelligence for detection and attribution of climate extremes*, XAIDA project, ICTP, Trieste, Italy
- **School** - *Machine Learning and the Physics of Climate* and Workshop on *Extreme Events in Weather and Climate*, Fletcher Hotel, Berg en Daal, The Netherlands
- EDIPI ITN 1st & 2nd training weeks - *Basics of statistics, probability, extreme events modelling and extreme value theory*, Uppsala University, Uppsala, Sweden
- EDIPI ITN 3rd training week - *Attribution studies, tipping point of climate and advanced statistical methods for extremes*, Ecole Normale Supérieure, Paris, France
- Journées de la physique statistique, Ecole Normale Supérieure, Paris, France
- Journées du lancement du Group De Recherche 'Défis théoriques pour les sciences du climat', Auditorium TOTEM, Paris, France

Talks and posters

- 2025 **University of Reading**, PROMOTE kickoff meeting. Presentation of the work: Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia.
- 2025 **EMS2025**, Presentation of the work: Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia.
- 2025 **University of Chicago**, Invited speaker. Presentation of the work: Gaussian framework and optimal projection of weather fields for prediction of extreme events.
- 2025 **Stormtracks2025**, Poster presentation of the work: Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia
- 2025 **ETH Zurich**, Invited speaker. Presentation of the work: Gaussian framework and optimal projection of weather fields for prediction of extreme events. Invited by Prof. Dr. Daniela Domeisen
- 2024 **European Geosciences Union General Assembly, EGU24**, Presentation of the work: A Gaussian Framework for Optimal Prediction of Extreme Heatwaves
- 2023 **GdR Defi théoriques pour les sciences du climat**, Poster presentation of the work: Heatwaves prediction using Gaussian assumption
- 2023 **Beg Rohu School of Out of equilibrium statistical physics**, Presentation of the work: Heatwaves prediction using Gaussian assumption
- 2023 **European Geosciences Union General Assembly, EGU23**, Presentation of the work: Comparing the influence of Atlantic Multidecadal Variability and spring soil moisture on European summer heatwaves

- 2022 **Tel-Aviv University, Department of Geoscience**, Presentation of the work: Comparing the influence of Atlantic Multidecadal Variability and spring soil moisture on European summer heatwaves
- 2022 **GdR Defi théoriques pour les sciences du climat**, Poster presentation of the work: Comparing the influence of Atlantic Multidecadal Variability and spring soil moisture on European summer heatwaves

Skills

Programming Python, MATLAB, C, Bash, L^AT_EX, Microsoft Office

Languages Italian, English, French, Neapolitan

Interests

Sports Running, hiking, swimming

Hobbies Baking, cooking, poetry and novels reading and writing, art's history and museums

Publications

- [1] Valeria Mascolo. *Studying extreme heatwaves using novel theoretical approaches*. PhD thesis, École normale supérieure de Lyon, 2024. <https://theses.hal.science/tel-04843914/>.
- [2] Valeria Mascolo, Clément Le Priol, Fabio D'Andrea, and Freddy Bouchet. Comparing the influence of atlantic multidecadal variability and spring soil moisture on european summer heat waves. *Oxford Open Climate Change*, 5(1), 2025. <https://doi.org/10.1093/oxfclm/kgae023>.
- [3] Valeria Mascolo, Alessandro Lovo, Corentin Herbert, and Freddy Bouchet. Gaussian framework and optimal projection of weather fields for prediction of extreme events. *Journal of Advances in Modeling Earth Systems*, 17(6), 2025. <https://doi.org/10.1029/2024MS004487>.
- [4] Valeria Mascolo, Francesco Ragone, Nili Harnik, and Freddy Bouchet. Rare events algorithm study of extreme double jet summers and their connection to heatwaves over Eurasia, 2025. <https://arxiv.org/abs/2507.13101>, *Submitted to Journal of Climate*.

References

Prof. Freddy Bouchet

- Address: Laboratoire de Météorologie Dynamique CNRS and ENS/PSL, LMD/IPSL and département de géosciences, 24 rue Lhomond, Paris, FR
- Contact: freddy.bouchet@lmd.ipsl.fr

Prof. Nili Harnik

- Address: Geophysics department, Kaplun 329, Tel Aviv University, Tel Aviv, Israel
- Contact: harnik@tauex.tau.ac.il , +972-3-640-6359

Prof. Corentin Herbert

- Address: Laboratoire de Physique, ENS de Lyon, 46, allée d'Italie, Lyon, FR
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Prof. Francesco Ragone

- Address: School of Computing and Mathematical Sciences, University of Leicester, LE17RH Leicester, United Kingdom
- Contact: fr120@leicester.ac.uk